SUBMISSION REPORT

eSim 2.5 — Ubuntu 25.04 Installation & Compatibility Fixes

Technical Debugging, Fixes, Verification & Open-Source Contribution

Candidate	Ahmed Umar Tahir Husain Pinjari
Primary Repository	Ahmed-Umar-Pinjari/eSim-Task4
Contribution Repository	Ahmed-Umar-Pinjari/eSim
Target Environment	Ubuntu 25.04 / eSim 2.5
Report Purpose	Final technical submission and reviewer-oriented evidence summary

Prepared from the current public GitHub repository state and documented project evidence.

1. Executive Summary

This report presents a structured account of the investigation, debugging, correction and verification performed for installing and running eSim 2.5 on Ubuntu 25.04. The work focuses on practical Linux and open-source software troubleshooting rather than only documenting a final successful installation.

The primary repository, eSim-Task4, preserves the troubleshooting trail through original and corrected installation scripts, logs, screenshots, issue-specific notes and a consolidated report. The current public repository describes multiple installation problems involving Ubuntu compatibility, libcanberra, GHDL archive paths, LLVM 18 compatibility, Verilator archive paths, NGHDL source paths, and Ngspice/NGHDL executable paths.

A separate eSim fork provides evidence of direct open-source contribution to FOSSEE/eSim. The contribution is tracked in upstream Pull Request #630, which is currently open and proposes restoring the missing Ubuntu NGHDL package used by the installer. The PR contains one commit and has not yet received a review; therefore this report does not claim that the contribution has been merged.

2. Submission Assets

Asset	Role	Current Evidence	Reviewer Value
eSim-Task4	Primary technical work	22 commits; fixes/, logs/, notes/, report/, screenshots/, README.md	Very High
eSim fork	Open-source contribution evidence	Public fork of FOSSEE/eSim	High
FOSSEE/eSim PR #630	Upstream contribution	Open PR restoring Ubuntu/nghdl.zip; 1 commit	Very High
Final report	Reviewer-facing explanation	This document	High

3. Objectives

- Investigate installation failures encountered while setting up eSim 2.5 on Ubuntu 25.04.
- Identify the technical root cause of each failure instead of applying undocumented trial-and-error changes.
- Create targeted fixes while preserving original scripts and before/after evidence.
- Verify the corrected installation using tool versions, executable paths, logs and screenshots.
- Organize the work so that another engineer can reproduce and audit the debugging process.
- Demonstrate practical contribution to the FOSSEE/eSim open-source project.

4. Target Environment

Component	Verified / Documented Value
Operating System	Ubuntu 25.04
eSim	eSim 2.5
Shell	Bash
Architecture	Linux
GHDL	4.1.0
LLVM	LLVM 18
Verilator	4.210
Ngspice	ngspice-35
NGHDL	Installed successfully

5. Debugging Methodology

The documented workflow follows a repeatable engineering sequence. The repository explicitly preserves the original failure state, investigates the source of the failure, applies a targeted correction, reruns the installation and records verification evidence.

1. Start with the original eSim installation.
2. Identify the installation error.
3. Capture the error using logs and screenshots.
4. Inspect the original installation script and relevant paths/dependencies.
5. Locate the missing, incorrect or incompatible component.
6. Create a backup/original version before modifying the script.
7. Apply a targeted fix.
8. Run the installation again.
9. Verify the individual component.
10. Record the resulting logs and screenshots.
11. Proceed to the next dependency or installation stage.
12. Perform final tool and executable-path verification.

6. Technical Issues Investigated and Resolved

Issue 1 — Ubuntu 25.04 Compatibility

Problem: The original eSim installation process encountered compatibility problems on Ubuntu 25.04.

Resolution: The original installation behaviour was captured before modifications, establishing a baseline for later comparison.

Evidence: Evidence is maintained in the repository's screenshots and logs.

Issue 2 — libcanberra Dependency

Problem: A missing libcanberra / related dependency prevented the installation from progressing normally.

Resolution: The dependency was investigated and the installation procedure was corrected for the Ubuntu 25.04 environment.

Evidence: The repository records both failure and successful verification evidence.

Issue 3 — GHDL Archive Path

Problem: The original installation function expected the GHDL archive at a path that did not match its actual location.

Resolution: The archive was found under the eSim source directory and the installation function was changed to use the correct source path.

Evidence: The documented extraction command is: tar -xJf "$src_dir/ghdl/$ghdl-source.tar.xz" -C "$HOME".

Issue 4 — LLVM 18 Compatibility

Problem: The Ubuntu 25.04 toolchain environment required a compatible LLVM version for the documented GHDL build process.

Resolution: LLVM 18 availability was verified and the GHDL source was configured/built using LLVM 18.

Evidence: The repository includes multiple screenshots and logs covering availability, installation and build verification.

Issue 5 — Verilator Archive Path

Problem: The Verilator installation initially failed because the expected archive was not available at the path used by the installation function.

Resolution: The archive location was investigated and the installation path was corrected.

Evidence: Final verification was performed using verilator --version, with Verilator 4.210 documented.

Issue 6 — NGHDL Source Path

Problem: NGHDL installation initially failed because the script could not locate nghdl-simulator-source.tar.xz.

Resolution: The NGHDL source archive location was investigated and the installation function was updated to use the correct source directory.

Evidence: The final NGHDL executable was verified at /usr/local/bin/nghdl.

Issue 7 — Ngspice / NGHDL Paths

Problem: The final environment required verification of Ngspice and NGHDL executable paths and their integration through soft links.

Resolution: The resulting installation was checked with which nghdl and which ngspice.

Evidence: Documented outputs are /usr/local/bin/nghdl and /usr/bin/ngspice respectively.

7. Final Verification

The repository's final verification section records the following environment state:

Tool	Final documented result	Verification method
GHDL	4.1.0	ghdl --version
Verilator	4.210	verilator --version
Ngspice	ngspice-35	ngspice -v
NGHDL	/usr/local/bin/nghdl	which nghdl

The documented final verification command sequence is:

```
echo "=== FINAL eSim TOOL VERIFICATION ==="
ghdl --version | head -n 3
verilator --version
ngspice -v | head -n 8
which nghdl
```

The repository reports successful installation of GHDL, Verilator, Ngspice and NGHDL, together with the required NGHDL/Ngspice path or soft-link verification.

8. Evidence and Reproducibility

- fixes/ — original, backup and corrected installation scripts used during the investigation.
- logs/ — original error logs, intermediate debugging outputs and final installation/verification logs.
- screenshots/ — visual evidence for original errors, fixes, installation stages and final verification.
- notes/ — issue-specific technical notes.

- report/ — consolidated installation issue documentation.
- README.md — reviewer-facing overview, environment, issue analysis, verification and repository structure.

This evidence-first structure is important because it allows a reviewer to distinguish the initial failure from the final state and to follow the reasoning behind each correction.

9. Open-Source Contribution to FOSSEE/eSim

The second repository is a public fork of FOSSEE/eSim. The current upstream Pull Request #630 is titled "Restore Ubuntu NGHDL package" and is authored by Ahmed-Umar-Pinjari.

According to the current PR description, the Ubuntu installer scripts reference Ubuntu/nghdl.zip during NGHDL installation, but the package was missing from the repository. The contribution restores the package without modifying the existing installer logic.

Contribution fact	Current status
Upstream project	FOSSEE/eSim
Pull Request	#630
Title	Restore Ubuntu NGHDL package
Target branch	FOSSEE/eSim: installers
Source branch	Ahmed-Umar-Pinjari: installers
Commits in PR	1
Review status	No reviews shown
Merge status	Open — do not claim merged

This is a valuable submission point because it demonstrates not only local debugging but also an attempt to return a concrete fix to the upstream open-source project.

10. Selection-Oriented Assessment

For a technical selection process, the strongest aspect of the submission is the combination of hands-on Linux debugging, shell scripting, source-build/toolchain compatibility, evidence management and an actual upstream open-source contribution.

Selection Dimension	Evidence	Assessment
Linux / Ubuntu debugging	Ubuntu 25.04 installation troubleshooting	Strong
Bash / scripting	Original and corrected installation scripts	Strong
Root-cause analysis	Archive paths, dependency and toolchain compatibility investigation	Strong
Verification discipline	Versions, executable paths, logs and screenshots	Strong
Documentation	Issue-wise README plus logs/screenshots/report	Strong
Git / version control	22-commit Task4 repository and separate contribution fork	Strong
Open-source contribution	Upstream FOSSEE/eSim PR #630	Strong
Reproducibility	Before/after scripts and preserved evidence	Strong

11. Key Strengths to Present During Selection

- I did not stop after getting an installation to work; I preserved the failure, investigation and verification trail.
- I used the actual filesystem/archive locations to identify path-related failures.
- I handled a toolchain compatibility issue involving GHDL and LLVM 18.
- I verified the final environment with explicit tool-version and executable-path commands.

- I maintained before/after script versions so the fixes can be audited.
- I organized evidence into logs and screenshots rather than relying only on a narrative README.
- I created an upstream contribution to FOSSEE/eSim for the missing Ubuntu NGHDL package.

12. Important Status Notes / Limitations

- The eSim-Task4 repository currently shows 22 commits and is public.
- The eSim fork is public and is explicitly marked as forked from FOSSEE/eSim.
- Upstream PR #630 is currently OPEN. It should be described as an submitted/open contribution, not as a merged contribution.
- The upstream PR currently shows no reviews. A reviewer may therefore evaluate the contribution based on the PR, commit and repository evidence.
- The final report should not claim that every eSim component or the entire application was rebuilt unless that is directly supported by the repository evidence.
- The strongest defensible claim is successful installation and verification of the documented eSim-related toolchain on Ubuntu 25.04.

13. Final Submission Checklist

- ☐ Open eSim-Task4 README on GitHub and manually open every log and screenshot link.
- ☐ Confirm every referenced file exists with exactly the same filename and capitalization.
- ☐ Confirm the repository is public and the final commit is pushed.
- ☐ Check git status locally and ensure no intended changes remain uncommitted.
- ☐ Confirm eSim fork is public.
- ☐ Keep the upstream PR #630 link available as contribution evidence.
- ☐ Do not claim PR #630 is merged unless GitHub shows it as merged.
- ☐ Submit both repository links in the task form exactly as requested.
- ☐ Use eSim-Task4 as the primary technical-work link and eSim/PR #630 as supporting open-source contribution evidence.
- ☐ Keep this report ready for interview questions about each issue, root cause, fix and verification command.

14. Reviewer Quick View

A reviewer can start with the following sequence:

13. README.md — understand the overall technical story.
14. fixes/ — inspect original and corrected installation scripts.
15. logs/ — verify the actual troubleshooting outputs.
16. screenshots/ — visually confirm major debugging and verification stages.
17. report/ — read the consolidated issue documentation.
18. FOSSEE/eSim PR #630 — inspect the upstream contribution.

15. Submission Links

Primary repository: https://github.com/Ahmed-Umar-Pinjari/eSim-Task4

eSim fork: https://github.com/Ahmed-Umar-Pinjari/eSim

Upstream contribution PR #630: https://github.com/FOSSEE/eSim/pull/630

16. Conclusion

The submission demonstrates a complete engineering workflow: reproduce the failure, collect evidence, inspect the installation process, identify the root cause, apply a targeted fix, rerun the affected stage and verify the result. The eSim-Task4 repository makes this workflow auditable through scripts, logs, screenshots and structured documentation.

The separate FOSSEE/eSim contribution strengthens the submission by showing an effort to address a real upstream installer dependency rather than keeping the solution only as a personal workaround. The most important final submission discipline is accuracy: present the Task4 work as the primary debugging project and present PR #630 as an open upstream contribution unless its status changes.